HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Engineered Python signal-processing pipelines for EMG data, implementing **advanced filtering** and **feature extraction** techniques to identify movement-associated patterns, achieving a **15% reduction** in noise artefacts.
- Developed and refined **EMG-based classification models** using **scikit-learn**, experimenting with alternative algorithms and feature engineering approaches to improve **gesture recognition** accuracy by **12%** on held-out datasets.
- Designed and executed reproducible testing workflows for EMG datasets, analysing classification errors and performance limitations to compare improvements against established baselines and produce evidence-based recommendations.
- Implemented **Python software solutions** for signal-processing and machine learning, writing clean, maintainable code and actively participating in technical discussions and experimentation to enhance platform capabilities.
- Investigated emerging developments in **EMG analysis** and **neural engineering**, translating research findings into practical software improvements for movement-intent recognition and future assistive technology applications.
- Managed **production support** for enterprise applications, responding to P1/P2 incidents and ensuring SLA adherence by analysing logs and monitoring dashboards in **Splunk** and **Dynatrace**.
- Containerised Python applications using **Docker** and deployed them to **AWS EC2** instances, configuring basic resource limits for predictable rollout and integration with existing CI/CD pipelines.

SKILLS

Signal Processing – EMG signal processing, Filtering, Segmentation, Feature extraction, Noise reduction, Time-series analysis

Machine Learning – Model training, Model evaluation, Classification algorithms, Gesture recognition, Movement-intent recognition, Scikit-learn, PyTorch, TensorFlow

Programming – Python, NumPy, SciPy, Pandas, Software development, Clean code, Documentation

Cloud & DevOps – AWS, EC2, Docker, CI/CD, Git, Jenkins

Data Analysis – Dataset analysis, Experimental validation, Performance measurement, Data interpretation

Domain Knowledge – Neural Engineering, Biomedical Signal Processing, Assistive technologies, Neurotechnology

Software Engineering – Code reviews, Technical discussions, Maintainable code, Software improvements

PROJECTS

EMG Signal Preprocessing and ML Model Optimization

- Developed a Python-based signal processing module to enhance **EMG data preprocessing**, applying custom **Butterworth filters** and implementing adaptive segmentation techniques that improved signal clarity by **20%**.
- Trained and evaluated multiple **machine learning models** (SVM, Random Forest) on processed EMG datasets for movement classification, achieving a peak **AUC of 0.88** on a public dataset for gesture recognition.
- Created automated testing scripts using **pytest** to validate signal processing steps and model performance, ensuring reproducibility and facilitating iterative improvements to the recognition pipeline.

Python Scripting for Data Analysis and Automation

- Automated data ingestion and cleaning workflows using **Python** and **pandas** for large-scale numerical datasets, reducing manual processing time by approximately **40%**.

- Developed **REST APIs** with **Flask** to serve processed data and model predictions, enabling integration with other internal systems and facilitating data access for downstream applications.
- Wrote comprehensive technical documentation and conducted knowledge-sharing sessions for new data processing scripts, improving team understanding and adoption of automated workflows.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)